SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Analytical Problem Solving

Course Code:	ITA0512	**Course Title:**	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	**Assessment:**	Assessment Tool 1 – **Analytical Problem Solving**
Weightage:	10% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*		
Student Name:	M.Manasa	**Reg. No.:**	192425189
Section / Batch:	AIML 2024	**Date:**	07/08/26

Code of Conduct

I __M.Manasa(192425189) __(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __m.manasa________________

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Criteria	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	Problem Understanding	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	Method Selection	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	Solution Process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	Accuracy of Calculation	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	Interpretation of Results	

7. Thresholding

(a) Explain global thresholding in segmentation.
(b) Given pixel values:
Apply threshold . Assign values (0 or 1).

(a) Explain Global Thresholding in Segmentation
Global Thresholding is one of the simplest and most widely used image segmentation techniques. It separates an image into two regions—foreground (object) and background—using a single threshold value (T). Every pixel in the image is compared with this threshold value.

- If the pixel intensity is greater than or equal to the threshold (T), it is assigned the value 1 (white/foreground).
- If the pixel intensity is less than the threshold (T), it is assigned the value 0 (black/background).

The mathematical representation is:
where:

- f(x, y) = Original pixel intensity
- g(x, y) = Segmented (binary) image
- T = Threshold value

Working Procedure

1. Read the grayscale image.
2. Select a suitable threshold value.
3. Compare each pixel with the threshold.
4. Assign 1 (white) if the pixel is greater than or equal to the threshold.
5. Assign 0 (black) if the pixel is less than the threshold.
6. The output is a binary image containing only the object and background.

Advantages

- Very easy to implement.
- Fast and computationally inexpensive.
- Suitable for images with high contrast.
- Useful for document scanning and object detection.

Disadvantages

- Not suitable for images with uneven illumination.
- Performance decreases when object and background have similar intensities.

- A single threshold cannot segment complex images accurately.

Applications

- Medical image segmentation.
- Fingerprint recognition.
- OCR (Optical Character Recognition).
- Industrial quality inspection.
- Satellite image analysis.

(b) Apply Threshold

Algorithm

1. Choose threshold value **T**.
2. Compare every pixel with **T**.
3. If pixel ≥ T, assign **1**.
4. Otherwise assign **0**.

Example

Pixel Matrix
40 70 120
150 90 200
80 180 60
Threshold = **100**

Binary Output
0 0 1
1 0 1
0 1 0

8. Edge Detection

(a) Explain the concept of edge detection.
(b) Apply the Sobel operator (horizontal):
to:
Compute the gradient at the center.

(a) Explain the Concept of Edge Detection

Edge detection is an image processing technique used to identify the boundaries of objects in an image. An edge is a location where the image intensity changes sharply from one pixel to another.
For example:

- Black object on white background
- Boundary between two different objects
- Corners and outlines of shapes

Edge detection is an important preprocessing step because it reduces unnecessary image information while preserving structural details.

Working Principle
The algorithm calculates the intensity difference between neighboring pixels. A large change indicates the presence of an edge.

Common Edge Detection Operators

- Sobel Operator
- Prewitt Operator
- Roberts Operator
- Canny Edge Detector
- Laplacian Operator

Advantages

- Detects object boundaries accurately.
- Reduces image complexity.
- Useful for feature extraction.

Disadvantages

- Sensitive to image noise.
- Broken edges may occur.
- Performance depends on image quality.

Applications

- Face recognition.
- Medical imaging.
- Autonomous vehicles.
- Object tracking.
- Industrial inspection.

(b) Apply the Sobel Operator (Horizontal)
The Sobel operator measures the intensity gradient in the horizontal direction.
Horizontal Sobel Kernel:
-1 0 1

-2 0 2
-1 0 1

Steps

1. Place the Sobel kernel over the 3×3 image window.
2. Multiply corresponding elements.
3. Add all products.
4. The result is the horizontal gradient.

Example

Image Window

10	20	30
40	50	60
70	80	90

Calculation

=(-1×10)+(0×20)+(1×30)
+(-2×40)+(0×50)+(2×60)
+(-1×70)+(0×80)+(1×90)

=-10+0+30
-80+0+120
-70+0+90

=80

Gradient at the center = 80

A higher gradient value indicates a stronger edge.

9. Region-Based Segmentation

(a) Explain region growing in segmentation.
(b) Given pixel intensities:
If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

(a) Explain Region Growing in Segmentation

Region Growing is a segmentation technique that groups neighboring pixels having similar intensity values into a single region. The algorithm begins from a **seed pixel** and expands by including neighboring pixels that satisfy a similarity criterion.

Working Procedure

1. Select a seed pixel.
2. Compare neighboring pixels with the seed.
3. Calculate the intensity difference.
4. If the difference is less than or equal to the threshold, include the pixel in the region.

5. Continue until no additional neighboring pixels satisfy the condition.

Advantages

- Produces connected regions.
- Preserves object boundaries.
- Easy to understand and implement.
- Effective for homogeneous regions.

Disadvantages

- Sensitive to seed selection.
- Incorrect seed may produce wrong segmentation.
- Computationally slower for large images.

Applications

- Medical image analysis.
- Tumor detection.
- Satellite image segmentation.
- Object extraction.
- Industrial inspection.

(b) Seed = 50, Threshold Difference = 5
Rule
A pixel belongs to the region if:
Therefore, the acceptable pixel values are:
45, 46, 47, 48, 49,
50,
51, 52, 53, 54, 55

Example
Pixel Matrix
44 48 52
50 55 60
47 53 70
Seed = **50**
Threshold Difference = **5**
Comparison:

- 44 → Difference = 6 ❌
- 48 → Difference = 2 ✅
- 52 → Difference = 2 ✅

- 50 → Difference = 0 ✅
- 55 → Difference = 5 ✅
- 60 → Difference = 10 ❌
- 47 → Difference = 3 ✅
- 53 → Difference = 3 ✅
- 70 → Difference = 20 ❌

Region Pixels
48 52
50 55
47 53
These pixels belong to the same region because their intensity values are within **±5** of the seed value (50) and are connected to the seed.